



10,000 people per hour per direction of traffic, needs a metro system. But building one is an expensive business. India's first metro project in Kolkata took 22 years to construct a 17 km-stretch at 14 times the initial cost. It served as a dampener on future projects. But that changed after the first phase of 65 km of the Delhi Metro was wrapped up three years before its 10-year deadline at Rs 10,571 crore, minus any cost overruns. It has made other state governments want to implement the same system in cities like Mumbai, Bangalore, Ahmedabad, Hyderabad, Chennai, Kochi, Hazrat Nizamuddin and Kolkata.

The reasons for this are simple. The economic rate of return and benefits to society from a metro system are evident within five years of its inception. Phase one of the Delhi Metro has taken 39,600 vehicles off the roads, slashed the consumption of petrol, diesel and CNG by 57,800 tonne and saved Rs 288 crore on road maintenance and traffic management. The aim is to bring a metro station within half a kilometre of each person's residence and workplace, which will discourage the use of private vehicles. When the system is supplemented by feeder buses, a light rapid transport system and a bus rapid transport system will make public transportation the first choice to navigate the city. The Delhi Metro is now carrying 0.6 million passengers a day, its punctuality helping people save 66 minutes of travel time a day.

Metro projects in other cities though are already running behind schedule. Some are following the build-operate-transfer model, where a private partner undertakes the construction and operation of the metro system for a certain time period before control reverts to the government. The public private participation model is the other option, where the Government makes the investment, and a private partner handles the operations.

But the ideal blueprint for a system of such magnitude is already in place in the Capital, where the Centre and state government work in collaboration. Concessions like land acquisition at low rates and a provision for electricity on a no-profit, no-loss basis have helped see the project through successfully. Replicating the system with an experienced technocrat heading the project in other cities, instead of administrative figureheads, is the way forward. India also has to construct a large reserve of technically competent people to build and run the system rather than import both technology and the human input.



Lessons from past:
The Kolkata Metro project served as a dampener on future enterprises as it took 22 years to construct a 17-km stretch at 14 times the initial cost

Working towards this, the Delhi Metro has tied up with the Indian Institute of Technology from 2008 for a specialised year-long course in metro technology.

Resources are another big hurdle, for which state governments should plan and raise dedicated funds. The usual passage for funds is to look for a private partner, take a loan or wait for dole from the Central Government. Instead, a game plan should be put in place, which can include high-parking fees, surcharge on fuel and higher registration fees for new vehicles.

Standardisation in terms of the system installed is the next step. It will not only cut costs but lead to the emergence of indigenous vendors and manufacturers who will assemble and supply parts for the metro. The first factory supporting metro projects will be functional in six months in Vadodara.

But beyond the economics, DMRC has achieved success due to a strong value system. It has been credited with following a work culture setting it apart from other government bodies by focusing on transparency, professional knowledge and punctuality. Re-inculcating a forgotten value system is the only way to root out corruption. This is the objective of the Foundation for Restoration of National Values, an association formed to create a framework to bring back a respect for values. It's a decisive step, which reflects on the success of every institution and project in the country.

The author is the managing director of the Delhi Metro Rail Corporation.

39,600
VEHICLES HAVE GONE OFF THE ROAD AFTER PHASE ONE OF THE DELHI METRO, SAVING RS 288 CR ON MAINTAINING ROADS AND MANAGING TRAFFIC.